CIT 140 Systems Analyst Portfolio

Portfolio Review Packet - Weeks 1 to 7

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CIT 140 - Introduction to Systems Analysis and Design

Week 8 - Portfolio Review - 4 November 2026

Repository: https://github.com/stuffthings15/CIT140-Workspace

All cases in this packet are fictional course material. No real machine, patient, facility or person is described.

Client overview

This packet collects seven weeks of systems analysis work and explains it for a reader who did not take part in the investigations. Each section names the question that was examined, the finding the supplied records support, why that finding matters, and what remains uncertain.

Two separate fictional cases run through the course and are kept apart deliberately. Meridian Forge Robotics is a robotics quality-assurance setting where a retired test began running again on a machine identified as MR-17; it covers Weeks 1, 2, 3 and 7. Cedar Junction is a civic operations and clinic setting; it covers Weeks 4, 5 and 6. No finding from one case should be read against the other.

Throughout this packet, evidence and proposal are kept distinct. A finding states what a supplied record shows. A proposal states what I recommend be checked or decided next, and is always attributed as such. Where a record does not establish something, that is said plainly rather than filled in with a reasonable guess.

Terms used in this packet

Artifact - the single deliverable produced in a given week, such as a brief, a model or a report.

Source record - a supplied case document, cited by its identifier, for example W05-S04 or W07-S02.

Flag - a marker that a record needs checking. A flag does not establish that a fault exists or that equipment is unsafe.

Swimlane diagram - a process diagram with one lane per role, showing who performs each step.

Data-flow diagram - a diagram of where information moves between people, processes and stored records.

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Every artifact named here is stored in the repository listed on the cover. The evidence list at the end gives the exact folder and file name for each one, and records the items that are missing.

1. Week 1- Problem Brief- Meridian Forge

Question examined

What happened to MR-17 on the night of the incident, and what do the supplied records actually support?

Supported finding

At 02:13:07 MR-17 accepted a command recorded as TEST_K44_START and at 02:13:09 began repeating Test K-44, a test the maintenance archive shows was retired 186 days earlier and removed from the active workflow. At 02:14:02 an operator sent a stop request. The robot received it and answered ACTIVE_INSTRUCTION_PENDING, and the test continued. The current scheduling console shows no active test assigned to MR-17 and no entry for K-44 in its catalog. Sources: the incident log kept by Nia Voss, the current scheduling console, and the maintenance archive.

Why it matters to this client

Production was paused in the affected quality-assurance cell. Beyond the stoppage, two controls the site relies on stopped being trustworthy at the same time: the scheduler no longer describes what the robot is actually doing, and an accepted operator stop request did not stop it. Those are separate problems from whatever caused the command.

Uncertainty and next check

The cause is not established. Command telemetry carries a source label reading legacy_sync.exe, but no supplied record identifies who owns or runs that service, and a label on a record is not proof of an actor. The next check is to trace that service, its owner, its trigger and its logs from around 02:13. A controlled restart was proposed by engineering but is not a verified fix, and it would clear the robot state that the investigation still needs.

2. Week 2- Business Case Recommendation- Meridian Forge

Question examined

Is the proposed QA Control Review worth authorizing, and on what terms?

Supported finding

My recommendation is to revise the request before it goes to the sponsor. The base estimate is 48 staff hours at 2,820 dollars, which fits the sponsor planning envelope of 3,000 dollars with 180 dollars to spare, and the eight business day plan fits a ten day limit. The preliminary planning range, however, reaches 3,525 dollars, which is 525 dollars above the envelope, and it excludes equipment, licences and overtime. Sources: the estimate and success-measure sheet W02-S05 and the request W02-S01.

Why it matters to this client

The deciding factor is not the arithmetic but a dependency. Capacity record W02-S04 states that read-only export access has been requested and not confirmed, and that if access is denied the work must be revised or deferred. Authorizing work whose central input may not arrive commits staff time against an unresolved condition.

Uncertainty and next check

No measured saving exists. Record W02-S02 supplies a baseline only: six retrieval requests totalling 60 staff minutes, of which two returned both an identified custodian and a link to the original source. Any claim that the review will reduce retrieval time is a target, not evidence. Before authorization, export access and the provisional staff hours should be confirmed, and the sponsor should either extend the envelope or cap the work and state what is dropped.

3. Week 3- Stakeholder Map- Meridian Forge

Question examined

Who is affected by the proposed review, who advises on it, and who can actually decide?

Supported finding

Three participants are named and their limits are documented. Nia Voss supervises night quality assurance and can confirm whether a record-checking workflow fits the work, but cannot approve funding. Elias Chen interprets command and scheduler records and can advise on technical feasibility, but record W03-S02 states that technical expertise is not proof that he owns the records or can authorize spending. Mara Ilyan routes recommendations to the sponsor but is not established as the spending approver.

Why it matters to this client

Two roles that the work depends on have no name attached. The participation and authority register W03-S03 records that the record-custodian entry has no named contact and no confirmation that anyone accepted the responsibility, and that the packet does not identify the person holding the sponsor role, their approval limits, or any signed approval. A night-shift operator, who performs the checks and records exceptions, has not been selected or interviewed.

Uncertainty and next check

A missing entry is a follow-up question, not proof that no one in the organization holds the responsibility. A contact appearing on a meeting list is not evidence of consent or authority. A timing conflict is also on record and unresolved: Operations proposed a single daytime meeting while quality assurance asked for night-shift participation, and W03-S04 records no agreement. Operations should identify the custodian and the sponsor and their limits before detailed requirements work begins.

4. Week 4- Validated Requirements Set- Cedar Junction

Question examined

What must the clinic appointment system do, stated so that each requirement can be checked?

Supported finding

Eight requirements were written and validated, each with a named source, an acceptance check and the stakeholder whose authority covers it. The central defect they address is a terminology collision recorded in W04-S01: scheduling uses the word held for a slot awaiting final confirmation, while intake uses the same word for a visitor who has arrived but cannot move forward. One label was carrying two different conditions.

Why it matters to this client

A visitor arrived holding a confirmation for 9:30 and was stopped at the desk because the intake queue displayed the appointment as held rather than ready. The requirements separate the two statuses, require the exception view to show which condition applies and who last changed it, and require a same-day slot to be offerable only when both a room and a qualified clinician are recorded as available.

Uncertainty and next check

Each requirement was confirmed only by the role whose authority covers it, and those limits are recorded rather than blurred: intake workflow by Dana Ruiz, capacity and readiness rules by Dr. Lena Okafor, access and communication needs by Priya Shah, and access, audit and availability constraints by Jordan Kim. None of them approves a technical implementation. The requirements state what the system must do; they do not choose a vendor or a screen design.

5. Week 5- As-Is Process Model- Cedar Junction

Question examined

How does the clinic appointment process work today, and where does the handoff between departments become unclear?

Supported finding

The current process was modelled as a swimlane diagram with four lanes: Visitor, Scheduling, Clinical Operations and Intake. Scheduling records the requested slot and support preferences, checks whether booking details are complete, and requests a clinical readiness decision. Clinical Operations checks room and clinician availability and either records readiness or retains a held status. Intake locates the appointment on arrival and checks the readiness status. Sources: the work notes W05-S01, W05-S02 and W05-S03.

Why it matters to this client

Drawing the lanes makes the break visible in a way a requirements list did not. The booking confirmation and the clinical readiness decision are two separate records, and nothing in the current process forces them to agree before the visitor arrives. In the incident trace W05-S04, a booking confirmation was issued while the slot was still held, so the visitor arrived holding a confirmation while intake saw an appointment that could not proceed. Record W05-S01 states directly that a booking message records the requested time and is not the clinical readiness record.

Uncertainty and next check

The supplied trace shows no completed readiness decision reaching intake before arrival, and W05-S04 states that missing from this trace does not prove an action never happened. It is therefore recorded as an open question rather than a finding of fault. The next check is the change history for the readiness decision: whether one was made and not recorded, recorded and not transmitted, and what triggered the confirmation while the slot was still held.

6. Week 6- Data-Flow Model- Cedar Junction

Question examined

Where does an inspection report go after it is accepted, and why did the department that needed one not receive it?

Supported finding

The Inspection Report Routing System was modelled as a context diagram and a first breakdown showing four processes, two stored records and three receiving departments. Comparing the inspection export W06-S02, the approved destination policy W06-S03 and the delivery log W06-S04 gives a clear result: five of eight reports reached the required department, and all three BRIDGE reports, CJ-603, CJ-604 and CJ-607, were delivered to Electrical instead of Structures. Structures received no work item at all in this export.

Why it matters to this client

Every report was validated as ACCEPTED, so the field-check step is not where this fails. The discrepancy appears after validation, where a destination is resolved. One of the misrouted reports was marked URGENT. The pattern tracks the asset type alone: it is not explained by priority, since one affected report was URGENT and two were ROUTINE and all three were routed identically, and it is not explained by survey area, since the three came from three different areas.

Uncertainty and next check

The cause is not established. Record W06-S03 states that it describes the required destination and does not prove what entries the running routing table contained, and no runtime snapshot, deployment history or code was supplied. W06-S04 states that a routing discrepancy is evidence of a result and that its technical cause remains a question. The next check is what entry the routing table holds for BRIDGE, when it last changed and by whom.

7. Week 7- Data-Quality Report- Meridian Forge

Question examined

Can the inspection records in the Meridian Forge tracking spreadsheet be trusted, and which records need checking?

Supported finding

Twenty records were prepared in Power Query, which trims spaces and converts status values to uppercase as a saved, repeatable step, and then checked against five rules from W07-S02. Eleven of the twenty records, 55 percent, fail at least one rule. Completeness Q01 flags 2 records, uniqueness Q02 flags 2, status validity Q03 flags 2, duration range Q04 flags 3, and the cross-field check Q05 flags 3.

The five rule counts total twelve while only eleven distinct records are affected, because record R03 fails two rules at once. It has no machine identifier and no inspector identifier while its status reads READY. Counting it twice would overstate how many records are affected.

Why it matters to this client

R03 states that an inspection is ready without identifying which machine was inspected or who inspected it, so nothing in that row can be traced to a physical asset or a person. The preparation step also matters: two records read as valid only after trimming and uppercasing, while a misspelled status and a status outside the agreed list remain flagged rather than being guessed at.

Uncertainty and next check

A flag means a record needs checking. It does not mean a machine is unsafe, and passing all five rules would not prove a record is accurate, because the checks test the record and not the machine. Source W07-S03 limits what this file can support. The verification routes are defined by role: the Maintenance Records Owner verifies a machine identifier or inspection time against the original inspection record, and the Shift Supervisor verifies a READY status or an inspector assignment.

8. Response to client review request W08-B01

The review request asks for one finding from the most recent case, the source or rule behind it, why it matters, and what should be checked before a design decision. The following responds to that request using my Week 7 data-quality work on Meridian Forge.

The finding

Record R03 in the inspection export fails two of the five data-quality rules in W07-S02. Rule Q01 requires that a machine identifier is present, and R03 has none. Rule Q05 requires that a record whose status reads READY identifies an inspector, and R03 has no inspector identifier while its status reads READY. The record therefore asserts that an inspection is ready without identifying either the machine inspected or the person who inspected it.

Why it matters

READY is the label the queue uses to move work forward, so this record can advance while remaining untraceable. Nothing in it connects to a physical asset or to an accountable person, which means a question raised about it later cannot be answered from the record itself. R03 is also the single reason the rule counts and the affected-record count differ, which matters to anyone reading the summary figures: twelve rule failures across eleven records, not twelve records.

What must be checked before any design decision

The first question is not how to prevent blank fields but why this record passed as READY in the first place. Before anyone designs a validation rule or a required-field control, I would ask: at what point in the current process is a status set to READY, and is any check applied at that point today? Without that, a new control could be added at a step that is not where the gap is.

The verification routes are already defined by role in W07-S03. The Maintenance Records Owner should verify the machine identifier for R03 against the original inspection record. The Shift Supervisor should verify the READY status and the inspector assignment. Neither value should be entered until the owning role confirms it, because a value typed in to clear a flag becomes indistinguishable from a verified value once saved.

I want to state one limit plainly, because the request raises it directly. A READY label and a passing format check describe the record, not the machine. Nothing in this spreadsheet establishes that any machine is physically safe or unsafe, and nothing in my Week 7 work should be read as a safety clearance.

9. Closing review and preparation for design

How the analysis has changed since Week 1

In Week 1 my instinct was to explain the incident. The supplied records named a service, legacy_sync.exe, and it was tempting to treat that label as the cause and move toward a fix. What changed across the seven weeks is the habit of separating what a record shows from what it would be reasonable to conclude. By Week 6 I was writing that a routing discrepancy is evidence of a result whose cause is still open, and by Week 7 I was recording an unverifiable field as a flag with a named owner rather than filling it in. The analysis got slower and more useful at the same time.

A second change is in how I handle people. Week 3 forced a distinction I had been blurring: responsibility is not authority. An engineer can be the only person who understands a record and still have no power to approve spending or to release access. Writing that distinction into every entry changed who I would ask about what, and it carried into Week 7, where each flagged field has a different verification owner.

Unfinished work

Stated honestly rather than left implicit. Every artifact each assignment required is present in the repository. One difference between the earlier and later weeks is worth explaining, because it looks like a gap and is not. For Weeks 2, 3 and 4 the assignment directed the mission receipt to Canvas rather than to the repository: those weeks asked for the artifact files to be committed and for the report link and receipt to be submitted in Canvas. From Week 5 onward the requirement changed and the receipt is stored with the artifact. The Week 2, 3 and 4 receipt codes are recorded in the repository README for completeness, taken from the original Canvas submissions, and no receipt has been reconstructed. Two items are genuinely open: Week 4 has no structured export in the repository, only the report, and the Week 4 link submitted in Canvas was a folder link on the main branch rather than the version-pinned file link that week asked for. Week 4 has no structured export in the repository, only the report. The folder layout is not uniform across the portfolio, because Weeks 1 to 4 use the structure each assignment specified at the time and those file links were already submitted. These are recorded in the repository README as well as here.

Three case questions also remain genuinely open rather than merely undone: what issued the retired test command to MR-17, what entry the Cedar Junction routing table holds for BRIDGE assets, and whether a clinical readiness decision for the stalled appointment was made and not recorded or recorded and not transmitted. In each case the supplied evidence establishes a result and stops short of a cause.

What I need to understand before comparing solutions or testing a design

Before comparing options I need the decision rights that Week 3 found undocumented. A comparison is only useful if someone can act on it, and at present the sponsor role is unnamed and the record custodian has no confirmed contact. I also need the access that Week 2 found unconfirmed, because several of the open questions can only be answered from records the team has requested and not yet received.

Before testing a design I need a baseline that can be measured again afterwards. Week 7 gives one: eleven of twenty records affected, with the specific rule failures identified. Week 2 gives another, the six-request retrieval sample, although it measures effort rather than outcome. What I do not yet have is agreement on which measure counts as success, and W02-S05 states that the sponsor must agree the target and the collection method. Without that, any later claim of improvement would be an assertion rather than a result.

No prototype or full requirements specification is proposed here. The portfolio is at the point where the problem, its value, its people, its requirements, its current process, its information flows and its data quality have each been examined, and the next step is comparing options against criteria that have not yet been agreed.

10. Evidence list and file locations

All paths are relative to the repository root at https://github.com/stuffthings15/CIT140-Workspace. Complete originals are preserved in the repository; this packet summarizes them and does not replace them.

Week	Case	Artifact	Location in repository	Receipt
1	Meridian Forge	Problem Brief (PDF)	Week01_Something_Isnt_Adding_Up/artifact/	In repo
1	Meridian Forge	Evidence packet (DOCX)	Week01_Something_Isnt_Adding_Up/evidence/	-
2	Meridian Forge	Business Case (PDF + JSON)	Week02_Is_This_Project_Worth_Doing/	Code on record; PDF missing
2	Meridian Forge	Repository check (PNG)	Week02_Is_This_Project_Worth_Doing/	-
3	Meridian Forge	Stakeholder Map (PDF)	Week03_Who_Has_A_Stake/	Code on record; PDF missing
3	Meridian Forge	Book form (PDF)	Week03_Who_Has_A_Stake/	-
4	Cedar Junction	Validated Requirements (PDF)	Week04_What_Must_The_System_Do/	Code on record; PDF missing
5	Cedar Junction	As-Is Process Model (PDF + drawio)	Week05_How_Does_The_Work_Happen_Now/Artifact/	In repo
5	Cedar Junction	Diagram image and sharing evidence (PNG)	Week05_How_Does_The_Work_Happen_Now/Evidence/	-
6	Cedar Junction	Data-Flow Model (PDF + drawio + PNG)	Week06_Where_Does_The_Information_Go/Artifact/	In repo
7	Meridian Forge	Data-Quality Report (PDF)	Week07_Can_We_Trust_The_Data/Artifact/	In repo
7	Meridian Forge	Workbook and Applied Steps (XLSX + PNG)	Week07_Can_We_Trust_The_Data/Artifact/	-
8	Both	This review packet (PDF + DOCX)	Week08_Your_Analyst_Portfolio/Artifact/	Week08_Your_Analyst_Portfolio/Receipt/